

A. Project Report (30%)

Criteria		PO	CP/ CA/ KP	Grad Attributes/ Taxonomy Level	Below Expectation	Between Acceptable to Outstanding	Outstanding	Score
General / Overall								
	Punctuality	PO8	WK7	GC4	Very late report submission	Late report submission	On-time report submission	1.5
	Clarity and conciseness	PO9	EA2	CS1	Main point/argument confused/ unclear. Irrelevant information. Unclear conclusion. Numerous spelling and punctuation errors. Severe grammatical error.	Argument reasonably clear; occasionally misses the point; not over-elaborate or over-complicated. Occasional lapses in spelling and punctuation. Some errors in grammar but not seriously distract the reader.	Argument effectively and efficiently conveyed; highly focused on the question; easily understood. Only few spelling and/or punctuation errors. Grammatically correct with complete sentences.	1.5
	Collaborative involvement	PO10	CA2	TW4	Focus of the project report is on one group member only	Not all group members have an active role in preparing the project report	All group members have an active role in preparing the project report	1.5
1	Abstract / Executive Summary							
	Abstract	PO9	EA2	CS1	Several major aspects of the report are missing, student displays a lack of understanding about how to write an abstract	Most of the major aspects of the report are pointed, some minor details are missing	Abstract contains all major aspects of the report, well-written	1.5
2	Introduction							

	Identification of Problem	PO2	WP6, WP4	TH6	Unable to identify and explain the specific problem that is going to be solved in the project	Somehow able to identify and explain the problem but the construction of logic is not really well structured	Able to identify and explain the problem and describe it in a well-structured logic	1.5
	Objective statement	PO9	EA3	CS1	Merely restating given objective; unclear objective	Adequate / clear objective	Well-written / precise objective	1.5
3	Method							
	Design strategy and creativity	PO3	CA5	C6	No design strategy, lack of effective innovation, haphazard approach	Use a design strategy with guidance and moderately effective innovation	Effective innovation with a systematic design strategy, including decomposition of work into subtasks, development of a timetable	1.5
	Integration	PO6	CP5	LS1	Unable to relate prior knowledge to the design problem. Has no concept of the process as a sum of its parts	Can use prior knowledge to design individual pieces of equipment competently when guided to do so. Does not think holistically	Understand how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem. Think holistically	1.5
	Modern Tool Usage	PO5	KP6	CTPS5	No application of engineering and/ or scientific principles	Applies engineering and/ or scientific principles incompletely	Applies engineering and/ or scientific principles correctly to design practical processes	1.5
	Methodology & evaluation	PO1	CP7	C6	Some methods or approaches were performed incorrectly or inappropriately; The usage of the method was not evaluated or was	The method or approach was used in a reasonable manner with few mistakes; The usage of the method was evaluated correctly using some relevant criteria	The method or approach was used appropriately, correctly and effectively; The usage of the method was evaluated correctly on all relevant criteria	1.5

					evaluated using inappropriate criteria			
4	Results and Discussion							
	Results	PO9		CS1	Figures, graphs, tables contain errors or are poorly constructed, having missing titles, captions or numbers, units missing or incorrect, etc	Most of the figures, graphs, tables are presented, but have some minor problems or could still be improved	All figures, graphs, tables are correctly drawn, are numbered and contain title/ captions	1.5
	Discussion	PO9		CTPS1	Very incomplete or incorrect interpretation of trends and comparison of data indicating a lack of understanding of results	Some of the results have been correctly interpreted and discussed; partial but incomplete understanding of results is still evident	All trends and data comparisons have been interpreted correctly and discussed, good understanding results are conveyed.	1.5
	Elaboration of Implementation Potentials and Limitations	PO12	KP7	ET3	No consideration of economic values, safety and environment	Includes only minor or cursory consideration of economic values, safety and environmental limitations	Discussion that includes economic, safety, environmental and other realistic limitations including commercial projections	1.5
	Solutions	PO4	CP4	C6	Only focuses on one solution to a problem; no optimization attempted	Can develop and compare multiple solutions to a problem; conducts optimization but neglects one or two key aspects	Development and justification of several potential solutions and finds optimum	1.5

	Practicality	P07	CA4	CTPS3	Design is incomplete, no answer is given	Give an answer, but does not check its practicality	Recognizes practical significance of design outcome/ answer	1.5
	Planning and Organizing	P011	KP5	CTPS4	Little or no attempt at strategic or merely gathering the data/ information; Minimal attempt to generate analysis in organized way	Effective use of strategic and good planning for analysis; Effective use of several methods to generate analysis in organized way	Outstanding use of strategic in critical way; well-planning for analysis; Outstanding use of various methods in creative way to generate organized analysis	1.5
	Cost estimation	P012		ES1	Not considering the estimation cost of the entire project	Inappropriately consider the estimation cost of the project	Appropriately consider the cost estimation of the project	1.5
5	Conclusion							
	Content	P09		CTPS2	Weak conclusion (not directly related to objective)	Conclusion partially supported by results/findings	Well supported conclusion based on findings. Conclusions are related to objectives	1.5
6	References							
	Number of reference	P011		LL1	< 10	10 - 20	> 20	1.5
	Academic integrity	P08	KP8	LL3	Work appears to be not adequately referenced or attributed	Other sources appear to be acknowledge	Appropriate use of others' work, acknowledged via in-text citations	1.5

**Project will be evaluated by IDP supervisors